
MAM1019H Hand-in 1 Semester 2 2022

Due: Friday 5 August 11:00 pm

Please make sure all of your answers are clearly labelled.

1. Determine whether each statement is true or false.

- (a) There exists a bijection $f : \mathbb{Q} \rightarrow \mathbb{R}$.
- (b) If $B \subseteq A$ and B is countably infinite, then A is countably infinite.
- (c) There is a subset A of $\mathbb{N}$ such that A is uncountable.
- (d) The set $\{0, 1\} \times \mathbb{N}$ is countable.

[4 marks]

2. Show that $\mathbb{N}$ and $A = \{n\pi : n \in \mathbb{N}\}$ have the same cardinality.

[3 marks]

3. Prove or disprove: the set of irrational numbers is countable.

[5 marks]

Maximum mark = 10

Available marks = 12